

Day 8: Probability & Statistics

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Outlines

- I. (Very old) lesson review
- II. Statistical inference
- III. Central limit theorem

1. The basis

Chance

Likelihood

Event

Occurrence

— — — — — — — — — —
Absolute

Symbolic maths

- A, B, C
- Ω
- $P(A)$
- A^c
 - $A \cup A^c = \Omega$
 - $A \cap A^c = \emptyset$
- (Ω, F, P)

Fill in the blank

- If A and B [...], we have $P(A + B) = P(A) + P(B)$
 - Else, $P(A + B) = P(A) + P(B) - P(AB)$
- If two events are [...], then $P(AB) = P(A)P(B)$
- A distribution describes the probabilities of events in general.

Expectations & variances

- $E[X] = \mu_X$
- $E[c] = c$, c is a number.
- $E[cX] = cE[X]$
- $E[X + Y] = E[X] + E[Y]$
- $Var(X) = \sigma_X^2 = E[(X - E[X])^2] = E[X^2] - (E[X])^2 = E[X^2] - E[X]^2$
- $\sigma_X \geq 0$

Conditional probability

- $P(A|B) \geq 0$ is the probability of A happening when B happens.
- Bayes: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- If all A are mutually exclusive:
 - $P(\sum_i A_i|B) = \sum_i P(A_i|B)$
 - $P(\Omega|B) = 1$
 - $P(AB) = P(A)P(B|A) = P(B)P(A|B)$
 - If $P(BC) > 0$ then $P(ABC) = P(C)P(B|C)P(A|BC)$

Inequalities

- Markov: $P(A \leq c) = \frac{E[A]}{c}$
- Chebychev: $P(|X - \mu| \geq d) \leq \frac{\sigma^2}{d^2}$
 - Simplified with $\alpha = \frac{d}{\sigma}$: $P\left(\frac{|X - \mu|}{\sigma} \geq \alpha\right) \leq \frac{1}{\alpha^2}$

2. Random variables (X)

- Objects that depends on random events.
- Two types: Continuous & Discrete.

Formulas

Discrete

- Prob. mass function (PMF)

$$p_X(d) = P(\{X = d\})$$

$$\sum_i p_X(d_i) = 1$$

Continuous

- Prob. density function (PDF)

$$\begin{aligned} p_X(c) &= P(\{\omega, X(\omega) \leq c\}) \\ &= P(\{X \leq c\}) \end{aligned}$$

$$\int_{-\infty}^{+\infty} p(x) dx = 1$$

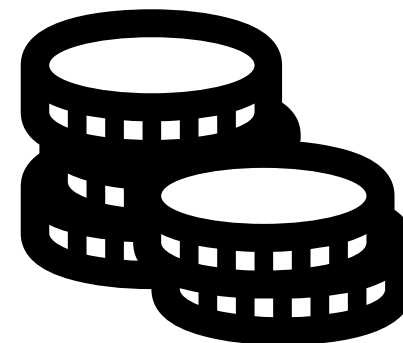
Distributions

Discrete

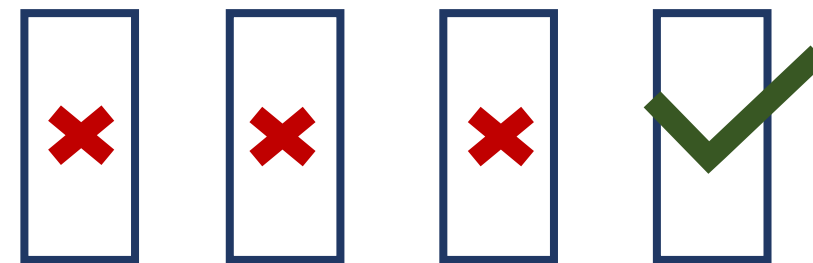
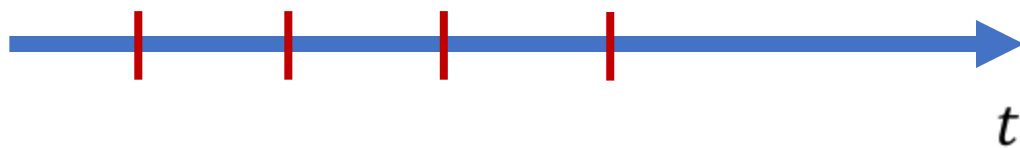
- Uniform
- Bernulli
- Binomial
- Negative binomial
- Hypergeometric
- Geometric
- Poisson

Continuous

- Continuous uniform
- Normal
- Standard normal
- Log-normal
- Gamma
- Cauchy/Lorentz
- Boltzmann
- Pareto



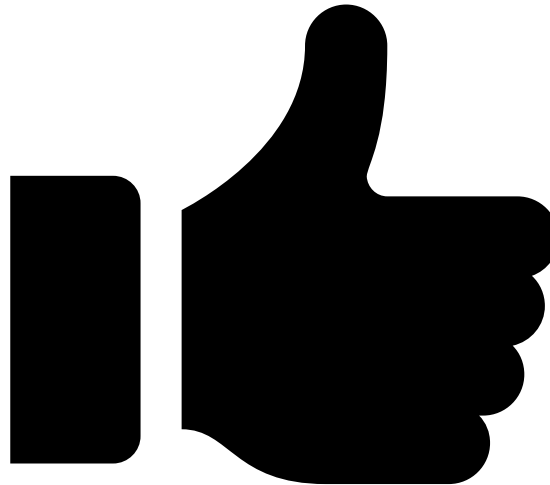
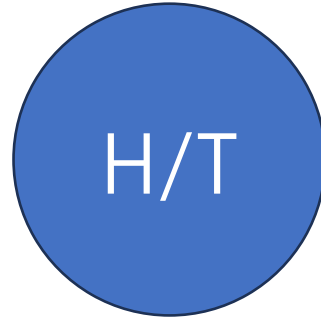
n events happen in t time

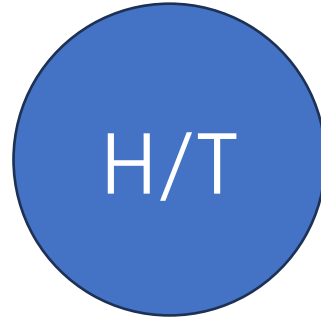


Experiments can be perfect

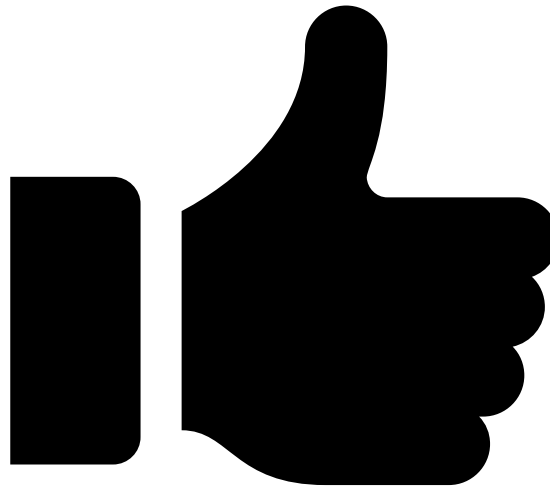
But mostly have sands

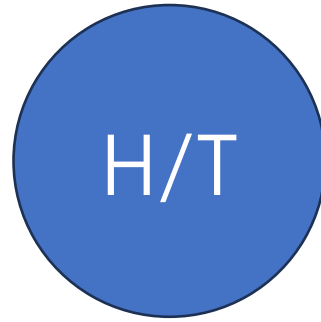
An example to make it clear



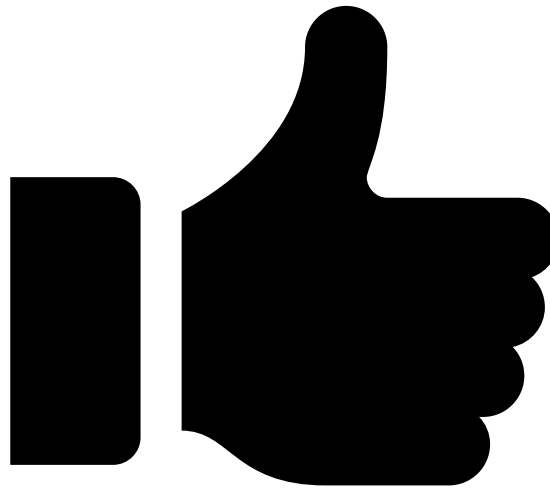


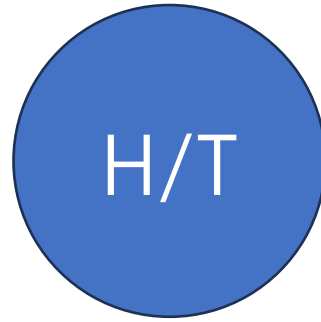
X100



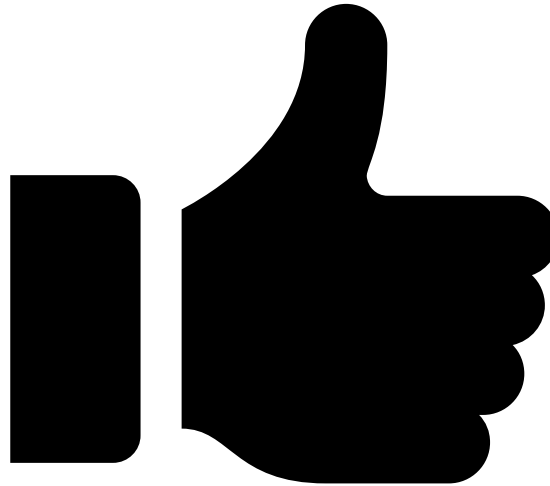


0.5/0.5





0.6/0.4



Ideally, we want to be near expected value

- It means 95 out of 100 experiments give predictable parameter in (a, b) called **confidence interval**.
- 0.05 chance of incorrectness is call α , or **significance level**.
 - Demonstrate **level of uncertainty**.
- 0.95 is the **confidence level**.

Formula

Margin of Error

$$CI = \bar{x} \pm \left(Z + \frac{\sigma}{\sqrt{|x|}} \right)$$

$\bar{x}$: Sample parameter

Z : Z-score

σ : Sample standard deviation

Confidence level	Confidence level
90%	1.645
95%	1.960
99%	2.576

Formula

Standard error

$$CI = \bar{x} \pm \left(Z + \frac{\sigma}{\sqrt{|x|}} \right)$$

$\bar{x}$: Sample parameter

Z : Z-score

σ : Sample standard deviation

Confidence level	Confidence level
90%	1.645
95%	1.960
99%	2.576

95% confidence interval interpretation

If there are $N \rightarrow \infty$ samples drawn from experiments of which there are exactly N intervals calculated, 95% of N intervals would successfully capture the true population parameter.

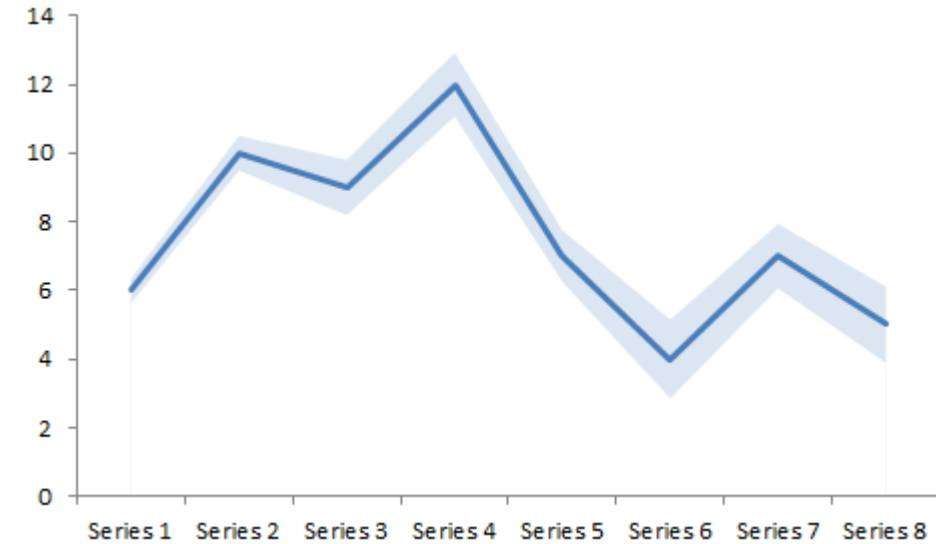
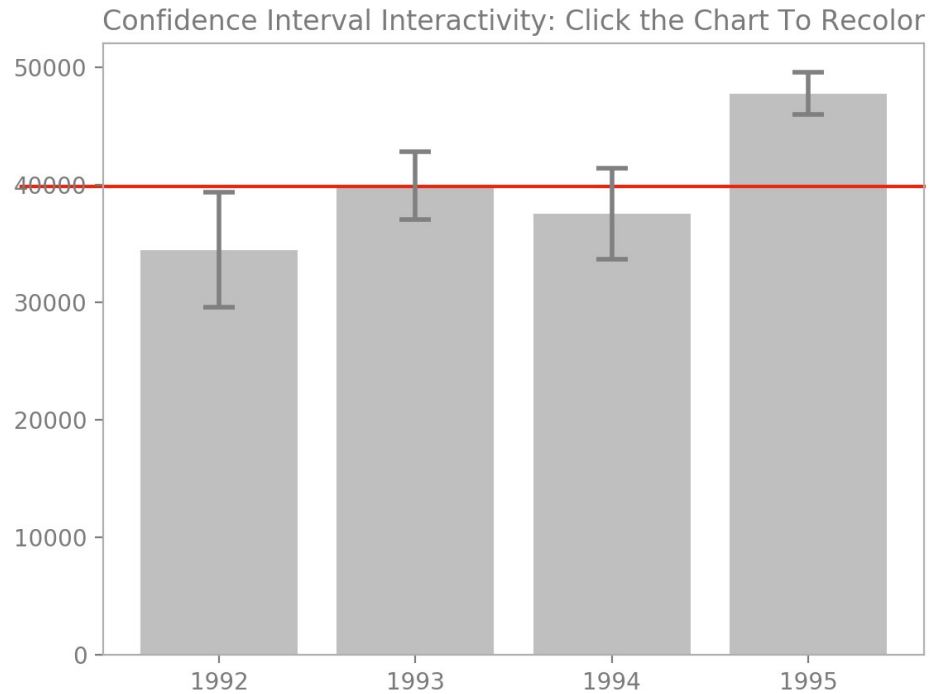
95% conf. interval (a, b) **does not** mean that...

- There's a 95% chance that the population params (mean, variance,...) fall into (a, b).
- 95% of data lie within (a, b).
- There's a 95% chance that the sample params (mean, variance,...) fall into (a, b).

After doing the experiments...

- We find the probability p that a trial is out-of-track.
- If $p \leq \alpha$, we reject the null hypothesis and accept our hypothesis.
- If $p > \alpha$, then be honest. 😞
- p is also called **p -value**.

Ok, so what's confidence interval in charts?



II. Statistical inference

- Relying on statistics techniques to find evidence for conclusions.
- Including:
 - Hypothesis testing (**Learned**)
 - Confidence interval (**Learned**)
 - Correlations (**Learned**)
 - Regressions (**Learned**)
 - Others: Chi-Square, contingency table, ANOVA, T-test.

Correlations

- In statistics, if random variable A correlate with B, then A linearly depends on B.
- It can be an evidence to prove causality.

We detected correlation by senses

- Causally correlated.
- One goes low, the other goes low/high.
- Ranking.

1, 2, 3, 7, 8, 9, 4, 5

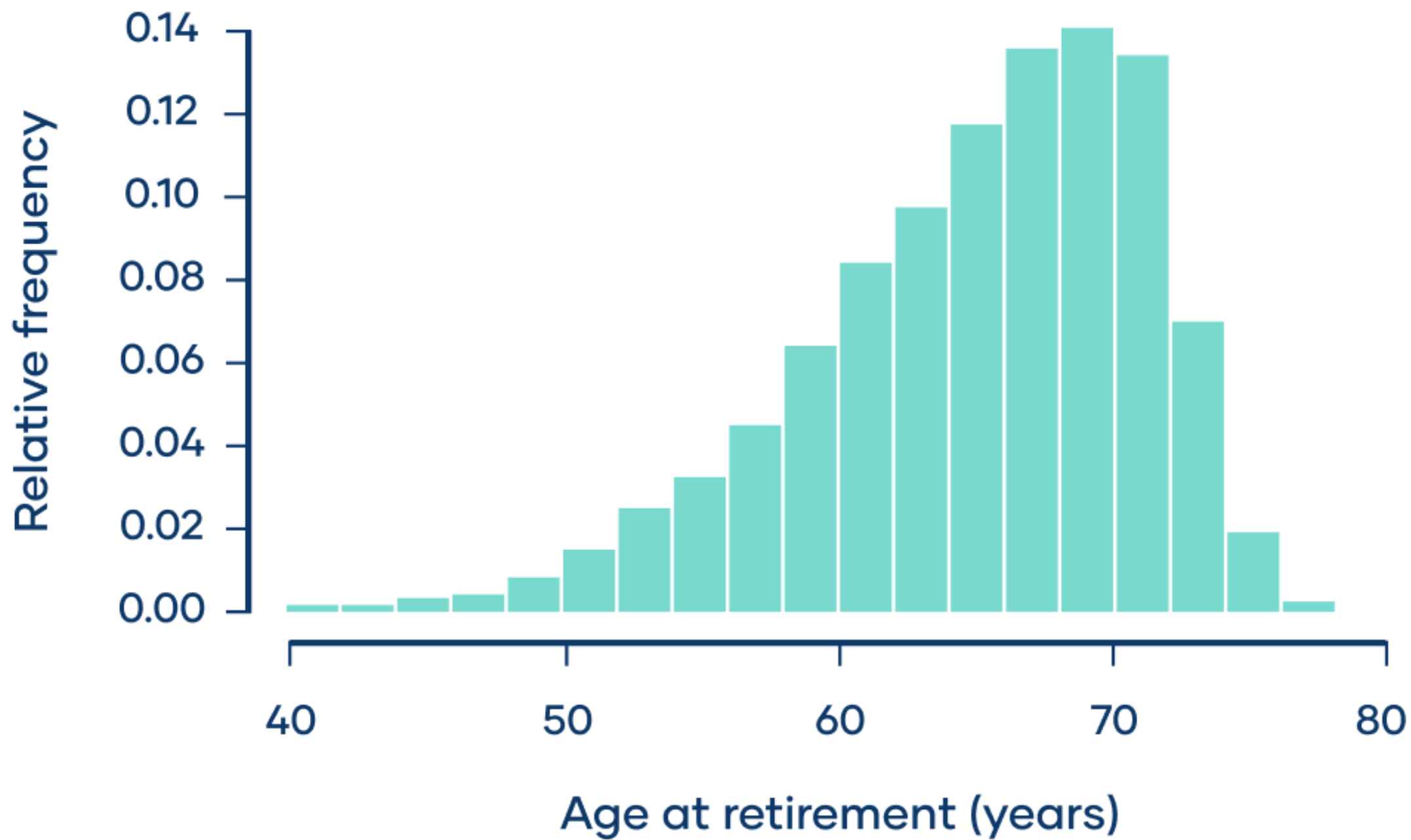
5, 4, 9, 8, 7, 3, 2, 1

Some metrics for correlations

- Pearson (linear, normal distribution)
- Point-biserial (linear)
- Kendall (rank-based, non-linear)
- Spearman (rank-based, non-linear)
- Cramér (nominal)

III. Central limit theorem

- If the samples are big enough, their means become normally distributed, regardless of the population distribution.
- It's called **sampling distribution**.



Trial #1

68, 73, 70, 62, 63

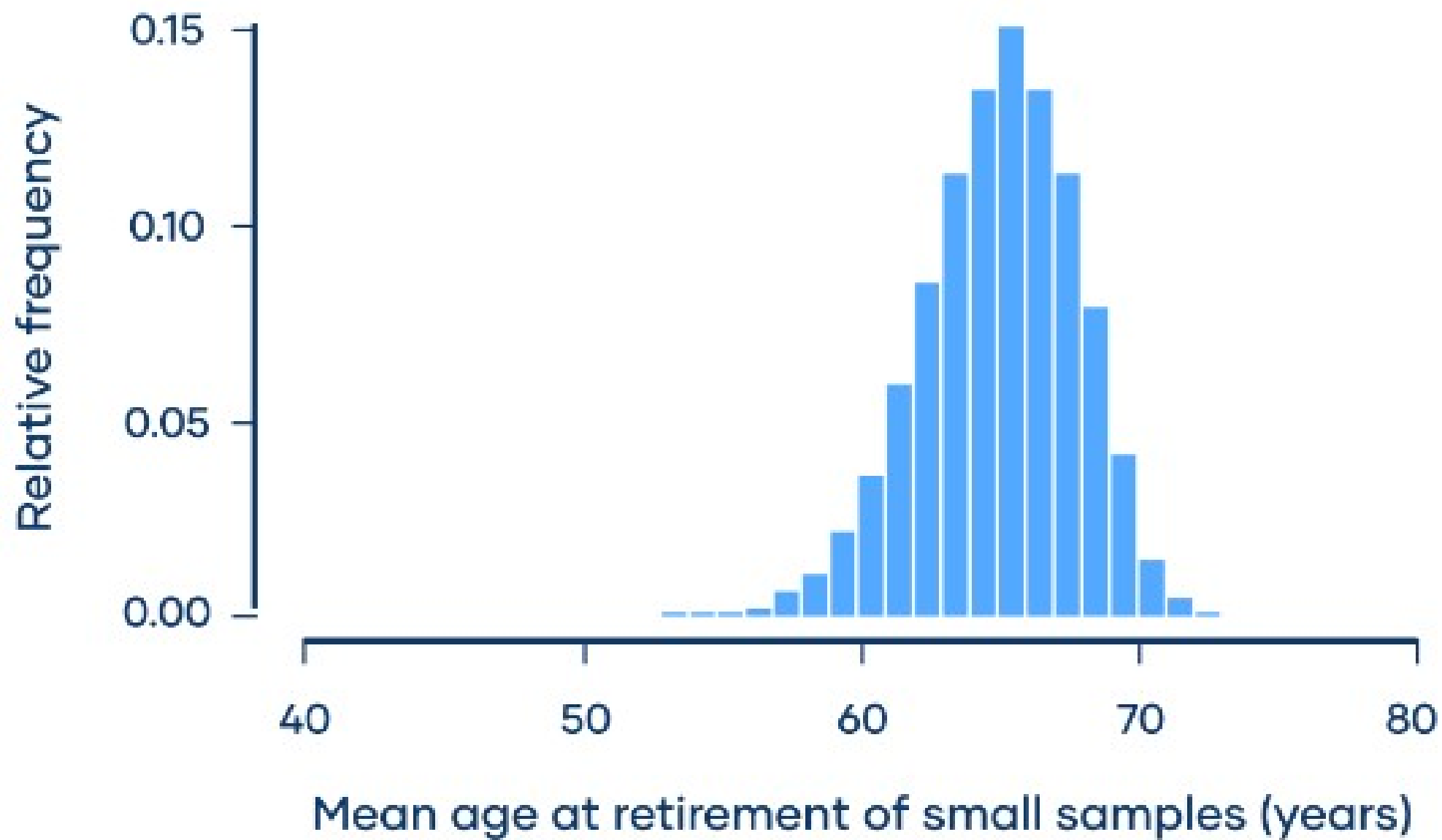
Trial #1

68, 73, 70, 62, 63

$$\mu_1 = 67.2$$

Trial #10

$$\mu = \{\mu_1, \dots, \mu_{10}\}$$



It's awesome to know
The data distribution.
But can we test
whether it's true or not?

Methods

- Graphical method
- Anderson-Darling statistic

If it's a normal distribution testing, we call it **normality test**.